

Foundation-Model-Based Retrieval-Augmented Generation (RAG) System using Hybrid Retrieval and Large Language Models

Project Overview:

In this project, you will develop a practical Retrieval-Augmented Generation (RAG) system that combines large language models with external knowledge retrieval techniques. The focus of the project is to build a realistic and implementable RAG pipeline while exploring modern retrieval and generation techniques.

Final Objective:

The final objective of this project is to design and evaluate a RAG framework that combines:

- Semantic document retrieval
- Foundation-model-based embeddings
- Large language model reasoning
- Hybrid retrieval techniques

You are expected to compare baseline and proposed methods, analyze retrieval quality, perform quantitative and qualitative evaluations, and submit a final IEEE-style report.

Recommended Project Scope:

You are strongly encouraged to focus on:

- Text-based RAG only (avoid multimodal/video RAG initially)
- One vector database (FAISS or ChromaDB)
- One or two embedding models
- One LLM integration
- Hybrid retrieval (Dense + BM25)
- Reranking or query expansion as the main contribution

Recommended Datasets for Standard Evaluation:

Dataset	Purpose	Difficulty	Recommended
SQuAD v2	Question answering and retrieval evaluation	Easy	Strongly Recommended
Natural Questions (NQ)	Open-domain QA retrieval	Medium	Recommended
MS MARCO	Passage retrieval benchmark	Medium	Strongly Recommended
HotpotQA	Multi-hop retrieval and reasoning	Hard	Optional
Custom PDF Dataset	Domain-specific RAG experiments	Easy	Recommended

Recommended initial evaluation setup:

- Use MS MARCO or SQuAD v2 for standard benchmarking
- Use a small custom PDF/document collection for domain-specific testing
- Limit dataset size initially for faster experimentation

Project Timeline and Tasks:

Phase	Timeline	Objectives	Tasks	Deliverables
Phase 1	Week 1–2	Understand RAG systems	Read papers and prepare environment	Literature summary
Phase 2	Week 3	Build retrieval pipeline	Implement semantic vector search	Baseline retrieval system
Phase 3	Week 4	Compare embeddings	Evaluate BGE, E5, SentenceTransformers	Embedding comparison
Phase 4	Week 5	LLM integration	Develop retrieval-augmented response generation	Initial RAG system
Phase 5	Week 6	Hybrid retrieval	Combine dense retrieval with BM25	Hybrid retrieval framework
Phase 6	Week 7	Optimization	Experiment with chunking and reranking	Optimization analysis
Phase 7	Week 8	Evaluation	Evaluate retrieval and response quality	Evaluation tables
Phase 8	Week 9	Analysis	Analyze failed retrieval cases	Analysis figures
Phase 9	Week 10	Final submission	Prepare report and presentation	Final deliverables

Recommended Technical Stack:

Component	Recommendation
Embedding Models	BGE-small, E5-base, SentenceTransformers
Vector Database	FAISS or ChromaDB
Framework	LangChain or LlamaIndex
Programming Language	Python
Retrieval Method	Dense retrieval + BM25 hybrid retrieval
Evaluation Metrics	Recall@K, Precision@K, MRR, Faithfulness

Suggested Research Contribution Ideas

Direction	Description	Difficulty	Recommended
Hybrid Retrieval	Combine vector search with BM25 retrieval	Medium	Strongly Recommended
Reranking Models	Use rerankers for improved retrieval ordering	Medium	Recommended
Query Expansion	Expand user queries using LLMs	Medium	Recommended
Chunk Optimization	Analyze chunk size and overlap impact	Easy	Strongly Recommended
Context Compression	Filter unnecessary retrieved information	Medium	Recommended

Expected Final Deliverables:

Deliverable	Description
Source Code	Well-organized implementation with documentation
Working RAG System	Retrieval and response generation pipeline
Experimental Results	Performance tables and comparison graphs
Visualization	Retrieval and embedding analysis
Final Presentation	Presentation slides
IEEE-Style Final Report	Technical report following IEEE format

Final Report Requirements (IEEE Style):

The final report must follow standard IEEE conference paper formatting and should include:

- Abstract
- Introduction
- Related Work
- Methodology
- Experimental Setup
- Results and Discussion
- Conclusion
- References

Recommended length: 6–8 pages in IEEE two-column format.